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Being internationally mobile while keeping domestic social capital: how postdocs from China manage precarity

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International academic mobility has surfaced as a new factor in the mechanism of cumulative advantages in early academic careers, yet its interplay with the role of doctoral mentors remains understudied compared to other factors. This research delves into the entanglement of international mobility and the doctoral mentor's role through a 4-year longitudinal study encompassing ten postdocs from a prestigious Chinese university's department. By employing narrative analysis across diverse cases, we found doctoral mentors wield multifaceted impacts on the access, decisions, experiences, trajectories, and outcomes of a postdoctoral international mobility through their role as teachers, sponsors, and collaborators. PhD graduates, supervised by eminent mentors with extensive overseas experiences, can enjoy a significant advantage in managing precarity and domestic social capital through mentors' impacts on international mobility. This study offers insights into the mechanisms underlying cumulative advantages fostered through international mobility, while revealing the enduring impact of doctoral mentors throughout the academic career development.

Keywords: cumulative advantages; doctoral mentor; international academic mobility; postdocs; academic career.

1. Introduction

The academic career of scientists is an ongoing process that can be characterized by different stages defined by significant events (Long and McGinnis 1985), all operating within the framework of the inequality mechanism of cumulative advantages. Among the crucial entry factors that initiate the process of cumulative advantages, early-career publication, PhD origin, doctoral mentor, gender, and ethnicity have been extensively studied (Allison and Stewart 1974; Bedeian et al. 2010; Gerhards, Hans, and Drewski 2018; Lindahl, Colliander, and Danell 2020). These factors act both independently and interactively (Long and McGinnis 1985; Cable and Murray 1999; Godechot 2016). However, much of the previous literature on the cumulative advantage mechanism has relied on cross-sectional data, limiting our ability to understand how specific entry factors function and interact over time. To address this limitation, in-depth longitudinal research with carefully selected cases is imperative.

In addition to the conventional factors, international academic mobility has emerged as a new determinant in the inequality mechanism of cumulative advantages in scientists' academic careers. International academic mobility can create or reinforce existing inequalities (Ackers 2008; Aceituno-Aceituno et al. 2017; Bilecen and Van Mol 2017), and consequently research interest has grown concerning the role of international mobility in the unequal process of cumulative advantages and its interactions with other entry factors. While current progress primarily focuses on the entanglement of international academic mobility with gender (Bilecen and Van Mol 2017; Schaer, Dahinden, and Toader 2017; Ivancheva, Lynch, and Keating 2019) and PhD origins (Gerhards, Hans, and Drewski 2018), its interaction with other crucial factors in academic careers remains understudied.

Recently, it has been found that young scholars' initial social capital, particularly in terms of academic networks and professional support, plays a crucial role in successful career development through the international academic mobility (Courtois and O'Keefe 2024). Therefore, it becomes specifically necessary to explore the interplay between international academic mobility and doctoral mentor's role. The postdoctoral stage, serving as the intermediary between doctoral education and the first stable academic appointment, offers an ideal opportunity for such analysis.

In many countries across Europe, Asia, and South America (Aceituno-Aceituno et al. 2017; Bilecen and Van Mol 2017; Sautier 2021), some PhD graduates strive to distinguish themselves from domestic counterparts through postdoctoral international mobility to compete for advanced domestic academic positions. However, to achieve this goal, they must navigate the challenges of academic precarity and maintain national social capital (Musselin 2004; Melin 2005; Sautier 2021; Schaer, Jacot, and Dahinden 2021; Courtois and O'Keefe 2024) beyond merely obtaining mobile opportunities. The sponsor role of doctoral mentors in facilitating mobile opportunities has been implicitly indicated (Kyvik, Karseth, and Blume 1999; Melin 2004), but a more comprehensive understanding is still needed on their influences on mentees' decisions, experiencess and outcomes of postdoctoral international mobility.

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Furthermore, most relevant research has been based on the Western European and North American contexts (Cantwell and Lee 2010; Stephan, Franzoni, and Scellato 2016; Bilecen and Van Mol 2017), while the value and accessibility of international academic mobility vary across cultures (Courtois and O'Keefe 2024). China, as the country conferring the largest number of doctorates, being a major source of international postdocs, and witnessing rapid development in its postdoc system (Cantwell 2011; Cyranoski et al. 2011; Song and Yang 2023), presents an intriguing case study, specifically when there are *Guanxi* and *Shimen* cultures in Chinese academia (Lu and McInerney 2016; Li and Tang 2019; Dai and Elliot 2022; Lai, Shi, and Lam 2023). However, little focus has been given to the early career development of domestically trained Chinese PhDs, their divergent trajectories of conducting postdoctoral research at home or abroad, and the doctoral mentors' role during this stage.

We try to address these research gaps through a longitudinal research of ten postdocs, who obtained their PhD degrees from the same department in a Chinese university. Through data collection from follow-up interviews, biographical information, and academic publication records, we tracked the postdocs' domestic and/or overseas postdoctoral career trajectories and subsequent employment status, with special attention paid to the role of doctoral mentors and postdoctoral international mobility. By employing narrative analysis of these multiple cases, we focused on the following research questions. First, how does the doctoral mentor's role interact with postdoctoral international mobility in postdocs' obtaining of cumulative advantages? Second, what are the consequences and influences of the intertwined doctoral mentor–international mobility?

Understanding these questions carries significant implications not only for a thriving global postdoctoral community but also for comprehending the implicit and long-term influences of mobility imperative in academic careers. On a theoretical level, our study enriches the understanding of how the mechanism of cumulative advantages operates when international academic mobility interacts with the influence of doctoral mentors, and how doctoral mentors continuously exert their influence on their mentees' career development during the postdoctoral stage and beyond.

2. Literature review

2.1 Postdoctoral international mobility

Postdoctoral positions have become increasingly common, particularly in the natural science fields (Cantwell and Lee 2010; Su 2011; Ahmed et al. 2015; Herschberg, Benschop, and van den Brink 2018). In contrast to research centres, such as the USA and UK, science, technology, engineering, and math PhD graduates in many countries face divergent choices for pursuing postdoctoral research either abroad or domestically to pursue academic careers. This divergence is influenced by nation-specific university configurations (Musselin 2004).

Postdoctoral international mobility, as a specific type of international academic mobility, is generally perceived positively due to its potential advantages in scientific productivity and international academic networking (Netz, Hampel, and Aman 2020). In several European and Asian countries, international academic mobility is increasingly associated with opportunities for advanced academic positions and research grants, supported by national policies and institutional measures (Musselin 2004; Cantwell 2011; Kim, Kim, and Gao 2015; Li, Shen, and Xie 2021; Sautier 2021). Consequently, international mobility has become a kind of symbolic capital, making it an important means for gaining cumulative advantages in early academic careers (Aceituno-Aceituno et al. 2017; Sautier 2021). Therefore, international mobile postdocs from peripheral countries are often regarded as exceptional candidates in their home countries and are considered more accomplished than their domestic counterparts (Cantwell and Lee 2010; Ahmed et al. 2015).

On the other hand, pursuing postdoctoral research abroad is more akin to a high-risk, high-reward bet rather than a guaranteed pathway to successful domestic academic careers (Musselin 2004; Melin 2005; Cantwell and Lee 2010). The situation can be more challenging for postdocs from developing countries who might face exploitation due to culturally specific stereotypes (Cantwell and Lee 2010). Overseas postdocs often encounter 'three dangers', including fixed-term appointments, risky research topics, and reduced or weakened domestic connections, which may lead to delays in publishing and a decline in domestic social capital, ultimately impacting their job prospects in their home countries (Musselin 2004; Melin 2005; Herschberg, Benschop, and van den Brink 2018; Bauder 2020). In the Chinese labour market, maintaining domestic social capital is specifically crucial due to the significance of *Guanxi* culture (an intricate cultural system with reciprocal relationships) (Lu and McInerney 2016; Li and Tang 2019).

In contrast, non-migrant postdocs experience fewer significant environmental changes, asymmetric information, and decreased national social connections. While this may facilitate maintaining productivity, it comes at the cost of international networks and symbolic capital derived from international academic mobility (Sautier 2021). Although widening international networks is possible through short-term visits, academic conferences, or collaboration with senior academics (Hauss 2021; Schaer, Jacot, and Dahinden 2021), the extent to which high productivity compensates for the lack of long-term international mobility remains uncertain. Additionally, the identity and responsibilities of postdocs vary across cultures, influencing how time and tasks are allocated, which in turn may impact productivity (Song and Yang 2023).

Previous literature has paid attention to the precarious nature of postdoctoral international mobility (Musselin 2004; Melin 2005; Sautier 2021; Schaer, Jacot, and Dahinden 2021), but it remains unclear how some mobile postdocs navigate the risky postdoctoral research stage, maintain domestic social capital, and attain an advantageous status compared to their domestic counterparts. Exploring the role of mentors in this process is particularly relevant, as young scholars often begin building their domestic and international academic networks from their doctoral mentors (Long and McGinnis 1985; Van Balen et al. 2012).

2.2 The doctoral mentor's role and postdoctoral international mobility

Doctoral mentors play a crucial role throughout scientists' academic careers, exerting various subtle influences as teachers, sponsors, and collaborators (Long and McGinnis 1985; Van Balen et al. 2012; Wang, Guo, and Wu 2021). However,

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their specific impact on their mentees' postdoctoral stage and postdoctoral international mobility remains unclear.

Some existing research focuses on the doctoral mentors' influence on postdoctoral international opportunities for their mentees (Kyvik, Karseth, and Blume et al. 1999; Melin 2004), as some international postdocs are recruited through global sourcing within the existing networks. For instance, a survey on Swedish postdocs from various disciplines in 1999 revealed that only 18 per cent of them had no prior contact with their postdoctoral employer, and the precontact channels often included their mentors' networks (Melin 2004). In Nordic countries' natural science disciplines, most doctoral students arranged their international mobility through their mentors' networks (Kyvik, Karseth, and Blume et al. 1999). Such arrangements enable postdocs to inherit international networks from their doctoral mentors (Melin 2004), giving them a competitive advantage over peers who lack such connections.

In addition to facilitating postdoctoral access, the doctoral mentors can significantly influence their mentees' domestic academic placement at the end of the postdoctoral stage (Long 1978; Godechot 2016). This mentor-driven academic placement mechanism is intertwined with research achievements, where the mentors' performance significantly affects their mentees' achievements, and the mentors' eminence indicates a process of ascription (Reskin 1979). Although this ascription advantage exists across various disciplines (Cameron and Blackburn 1981; Long and McGinnis 1985), it is challenging to measure. Among a relevant literature, a natural experiment on faculty recruitment in French social sciences revealed that when the mentor was randomly involved in the electoral commission, the chance of the applicant being selected into the candidate list doubled (Godechot 2016).

However, the doctoral mentors' influences extend beyond entry and final point for the postdoctoral stage. Firstly, mentors can mobilize their social capital to ensure their mentees' job security, making the decision to pursue postdoctoral international mobility easier. Secondly, the inheritable international network can influence not only the opportunities but also the experiences of postdoctoral international mobility. Benefitting from the inherited international connections, some postdocs can gain access to richer information channels, be more likely to be trusted rather than exploited, and be less likely to lose the domestic connections associated with the mentor (Musselin 2004). Thirdly, as teachers and role models, doctoral mentors have the power to influence their mentees' career decisions by guiding them towards specific career paths (Long and McGinnis 1985; Van Balen et al. 2012). Additionally, some scholars expect their mentees to follow in their academic career trajectories to justify their past decisions, expand their network, and increase their academic influence (Blackburn, Chapman, and Cameron 1981), which may lead some PhD graduates to choose overseas postdoc positions if their mentors also have long-term overseas research experiences. Lastly, doctoral mentors can maintain collaboration with their mentees during the postdoctoral stage, particularly when the postdoctoral research and network are closely related.

The above-mentioned influences of doctoral mentors in domestic job security, inheritable international networks, career guidance, and research collaboration intertwine with the access, decisions, experiences, and outcomes of postdoctoral international mobility, but empirical evidence in this regard is still insufficient.

3. International mobility, doctoral mentor's role, and postdocs' cumulative advantages

The mechanism of cumulative advantages operates through the joint effects of multiple factors and specific contexts in an intertwined and reinforced manner (Long and McGinnis 1985; Merton 1988; Cable and Murray 1999; DiPrete and Eirich 2006; Godechot 2016). According to Turner's contest mobility norm and sponsored mobility norm, the general factors in cumulative advantages can be divided into performance dimension and ascription dimension, aligning with Merton's notions of universalism and particularism, respectively (Cable and Murray 1999). For instance, scholars' publication records are typically used to measure performance, while doctoral mentorship, PhD origin, gender, and race serve as ascription factors. According to the fundamental norm of universalism in science and academia, performance should be the primary criterion for allocating academic positions (Kim, Kim, and Gao 2015). Nevertheless, ascription factors, such as the prestige of the PhD origin and doctoral mentorship, can play a crucial role (Cable and Murray 1999; Cantwell 2011; Godechot 2016; Laufer 2020).

As a new determinant in academic careers, international mobility is not only legitimized through the concept of research excellence in academia, but it can also exacerbate inequities by reinforcing other ascription factors in cumulative advantages (Ackers 2008; Bilecen and Van Mol 2017). For instance, international academic mobility patterns exhibit distinct gender-specific characteristics, often aligning with traditional male stereotypes of the ideal academic entrepreneur (Leemann 2010; Bilecen and Van Mol 2017; Ivancheva, Lynch, and Keating 2019; Malakhov, Vasil'eva, and Belov 2020; Li, Shen, and Xie 2021). Furthermore, the hierarchical reputation of academic institutions significantly impacts candidates' chances for international mobility (Gerhards, Hans, and Drewski 2018). The doctoral mentor factor, considered as the first and vital step in the mechanism of cumulative advantages, may also increase sponsored mobility via their influences in mentees' international mobility (Lai, Shi, and Lam 2023).

Doctoral mentors' influences on international mobility can vary across different disciplinary backgrounds. Compared to social sciences, natural sciences, such as physics, biology, and chemistry, have a relatively high rate of postdoctoral international mobility (Cantwell 2011; Rostan and Ceravolo 2015; Li, Zuo, and Shen 2018; Malakhov, Vasil'eva and Belov 2020; Momeni et al. 2022; Xu and Shen 2024). Meanwhile, there is evidence that the doctoral students' international visits in these fields are often arranged by doctoral mentors, whereas in social sciences international visits are mainly arranged by students themselves (Kyvik, Karseth, and Blume et al. 1999; Lai, Shi, and Lam 2023).

Furthermore, doctoral mentors' influences on international mobility can also vary to different degrees in different cultural contexts. In China, the prevalent *Guanxi* culture, alongside the *Shimen* system, reinforces the pivotal role of doctoral mentors in shaping academic careers. *Shimen*, characterized as a semi-closed collective learning community cantered around a shared mentor, fosters mentor–mentee relationships and

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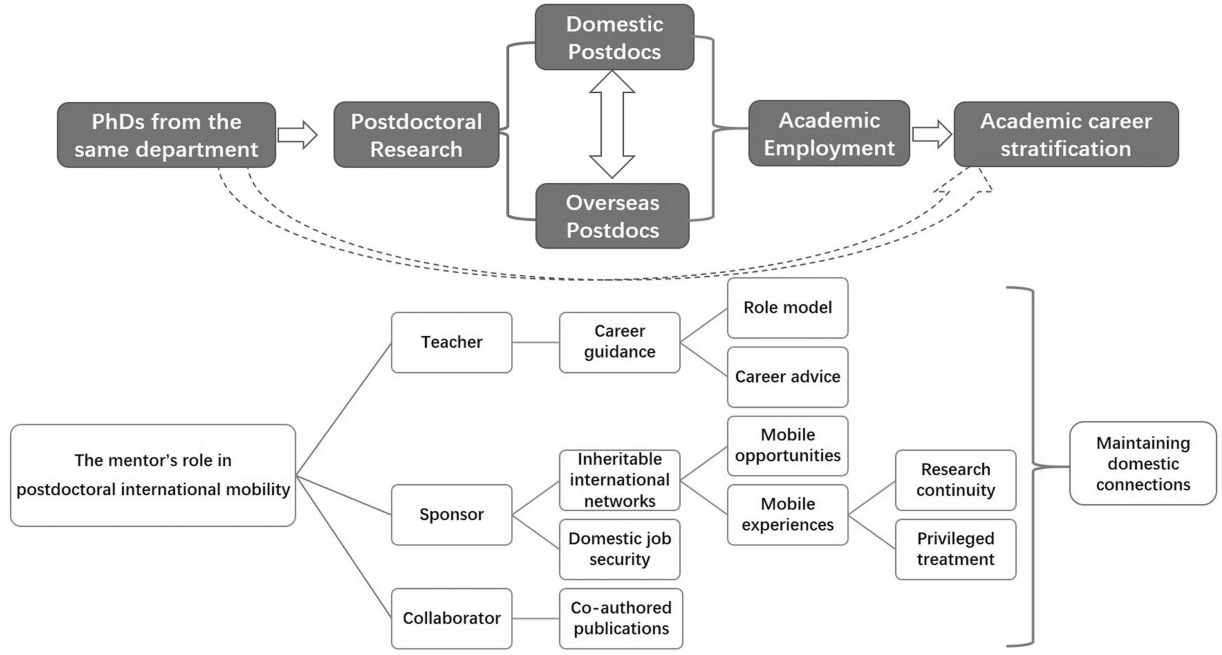

Figure 1. The doctoral mentor's role in mentees' postdoctoral international mobility.

connections, functioning as an informal organizational structure (Lin and Chao 2019; Dai and Elliot 2022). Recent observations reveal that Chinese academic leaders in physics often arrange their master students' international mobility, from pursuing overseas PhD degrees to returning as faculty members (Lai, Shi, and Lam 2023). This tacit approach enables mentors to accumulate more advantages within the *Shimen* community, yet potentially undermining academic independence and equality among budding scholars. In other cultures, such as Ireland and Switzerland, doctoral mentors may serve as professional support and initial social capital in mentees' postdoctoral international mobility as well (Leemann 2010; Courtois and O'Keefe 2024).

To fully examine the intertwined doctoral mentors' role and postdoctoral international mobility in cumulative advantages of early academic careers, we tracked the trajectories of domestic and overseas postdocs who obtained their PhDs from the same department. The tracking research commenced at the early stage of their postdoctoral research and concluded with an analysis of the stratification of their employment status. To investigate the doctoral mentor's influences on postdoctoral international mobility, we examined their role as a teacher, sponsor, and collaborator, as discussed earlier and illustrated in Fig. 1.

4. Research methods

Scientists' early careers can be influenced by a myriad of factors that are challenging to control (Van Balen et al. 2012). To address this challenge, Van Balen proposed a strategy of comparing cases with enough variety yet enough similarity (Van Balen et al. 2012). Following this strategy, our study focused on postdocs who embarked on their postdoctoral research during a similar timeframe, sharing identical disciplinary backgrounds and PhD institutions, yet differing in their mobility experiences and doctoral mentors.

Considering mobility patterns are embedded in specific research practice regarding discipline background, national and organizational measures (Musselin 2004; Laudel, Bielick, and Glaser 2019; Li, Shen, and Xie 2021; Courtois and O'Keefe 2024), samples were carefully selected in this research. In 2018, cases of PhD graduates were collected from a chemistry department in an elite Chinese university, as chemistry is a typical natural science discipline (Balaban and Klein 2006), and long-term experiences of international mobility was important in securing academic careers in this department. Focusing specifically on academic trajectories, we only invited those PhD graduates who had demonstrated exceptional performance and having received at least one academic award during their doctoral education. Out of the eighteen accessible candidates who met our criteria, ten chose to pursue postdoctoral research, with five opting to remain in China and the remaining five selecting overseas positions. All of them embarked on their postdoctoral journeys around 2016. To further examine the role of postdoctoral international mobility and the doctoral mentor factor, we maintained communication with these ten postdocs and followed their academic progress for 4 years (2018–2022), tracing their career trajectories and experiences. Follow-up interviews were conducted in 2021 on their postdoctoral research experiences, current career status, and future career aspirations. Their career status was updated in April, 2022. Table 1 presents basic information for the ten postdocs, with all names replaced by pseudonyms.

Regarding research performance, we retrieved all the participants' publication records from Web of Science (WoS) until 12 February 2022, as presented in Table 2. As academic motivation is also considered an indicator of talent excellence (Van Balen et al. 2012), we categorized the postdocs' academic motivation into three levels based on the first interview. Those who considered academic careers as their primary choice were marked with high motivation, while

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Table 1. Basic information of the interviewees.

Samples (pseudonym)	Gender	Postdoctoral research (location and time length)			Career status
Jin	M	USA, 3 years			An independent PI at an elite university of China
Han	F	USA, 3.5 years			An independent PI at an elite university of China
Wen	M	USA, 1 year	Unemployment, 0.5 year	USA, 3.5 years	A co-PI at an elite university of China
Lang	M	USA, 2 years			An industry job in the USA
Min	M	USA, 4 years			An independent PI at an elite university of China
San	F	China, 2 years			An industry job in China
Zheng	M	China, 2 years	USA, 1 year	USA, 2 years	An independent PI at an elite university of China
Yan	M	China, 2.5 years			An independent PI at a less prestigious university in China
Rong	F	China, 2 years			An industry job in China
Chen	M	China, 5 years and continuing			Remain a postdoc

Table 2. The interviewees' publications in different stages.

Samples (pseudonym)	Level of academic career tendency during doctoral education	Publications during doctor education		Publications during postdoctoral research		Total publications from doctoral to postdoctoral period	
		Sum of IF	Citations	Sum of IF	Citations	Sum of IF	Citations
Jin	Medium	73	646	30	298	103	944
Han	Medium	11	4	91	442	103	446
Wen	High	50	145	52	224	102	369
Lang	Medium	66	239	8	114	74	353
Min	High	38	191	16	146	54	337
San	Low	5	20	5	53	10	73
Zheng	Medium	55	318	China 49/43	USA 15/7	119	368
Yan	High	79	663	30	88	109	751
Rong	Medium	12	50	13	45	25	95
Ben	High	9	110	29	48	38	158

those who hesitated between academic and non-academic careers were marked with medium motivation. Postdocs who preferred non-academic careers were marked with low motivation.

For investigating the role of doctoral mentors, we collected biographical information on the mentors, as shown in Table 3. Special attention was paid to their publication records and honorary titles to reflect the mentors' performance and eminence (Reskin 1979). In the context of Chinese chemistry, honorary titles associated with the mentors' eminence included The National Science Fund for Distinguished Young Scholars, the Changjiang Scholar of the Ministry of Education, the Member of the Chinese Academy of Sciences, and the winner of National Natural Science Award. We also considered mentors' international honorary titles, such as being a foreign academician, a full member of the chemistry society of a foreign country, or holding an honorary title from the chemistry society of a foreign country. Additionally, we collected information on mentors' long-term full-time overseas experiences.

This research employs narrative analysis to present the research findings, delving into people's experiences and behaviours to understand the underlying meaning through storytelling (Clandinin and Connelly 2000). In our study, we conducted a similar cross-case analysis following the previously reported method (Van Balen et al. 2012), while also utilizing self-comparison in different stages through the longitudinal study to facilitate within-case analysis.

5. Research findings

Among the ten postdocs, three decided to leave academia due to a lack of research interest and/or the exhaustion of competitive academic life (referred to as San, Lang, and Rong). The remaining seven postdocs pursued academic careers, and six of them secured tenure-track or permanent positions in prestigious Chinese universities (as shown in Table 1). Based on their narratives, we identified three distinct academic trajectories: (1) going abroad for postdoctoral fellowships, (2) staying as domestic postdocs, and (3) combining advantages as domestic and overseas postdocs. In the following section, we present typical cases for each trajectory using both within-case and cross-case analysis, illustrating how certain postdocs achieved success in their early careers while others faced challenges, and how international mobility and mentor

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Table 3. The mentors' overseas experiences and honorary titles.

Samples (pseudonym)	Mentor's H index	Mentor's total citation	Mentor's overseas experiences (years)	Mentor's number of honorary titles	
				Chinese	International
Jin	118	63,634	7	3	1
Han	72	18,385	10	4	3
Wen	50	8,878	5	3	0
Lang	72	18,385	10	4	3
Min	30	2,749	8	2	1
San	40	5,802	9	1	0
Zheng	77	21,416	2	4	1
Yan	30	2,955	No long-term experiences	3	1
Rong	35	3,765	No long-term experiences	2	0
Ben	30	2,749	8	2	1

influence intersect to shape cumulative advantages during the postdoctoral stage.

5.1 The successful model of early career in academia: Jin and Han

Jin and Han emerged as the most successful postdocs in this study, obtaining tenure-track positions at esteemed Chinese universities at a rapid pace. They shared two key factors in common contributing to their success: an eminent doctoral mentor with extensive overseas experience, and an overseas postdoctoral position arranged within their mentor's close network.

Jin has impressive research performance, evidenced by predoctoral publications. His main motivation to pursue postdoctoral research in the USA was to obtain the symbolic capital of international mobility. He deliberately chose a world-leading organization where his doctoral mentor also conducted postdoctoral research, and his experiences markedly differed from those of migrant postdocs who were sometimes perceived as inexpensive labour rather than future faculty (Musselin 2004). Benefiting from the existing network, Jin enjoyed privileges in accessing resources, receiving high-quality training, and accumulating various academic capitals beyond his initial goal of gaining symbolic capital through international mobility. His case highlights how an inherited international network not only opens overseas opportunities but also provides privileged treatment during the whole postdoctoral stage:

> 'The postdoctoral experiences depend on how close the postdoctoral mentor is with your doctoral mentor… He (the postdoctoral mentor) is a good friend of my doctoral mentor, so it's natural that he always treats me as people on his own side, and assigns me various things…including but not limited to fund applications, mentoring doctoral students and a lot of social work with different countries.' (Jin, second interview)

Han's doctoral publication profile was not as outstanding as Jin's. Among the ten participants, her academic output and aspirations during her doctoral education were moderate. However, her mentor, who held a leading position in Chinese academia and had a well-connected international academic network, played a pivotal role in securing an international postdoctoral offer relevant to her previous research. This ascriptive advantage revitalized her academic aspirations as she engaged in 3.5 years of continuous postdoctoral research under the guidance of a productive Chinese scholar in the USA. Like Jin, her postdoctoral mentor treated her well, and she felt respected rather than exploited. In comparison to Rong and Ben, who exhibited similar productivity during their doctoral education, Han's postdoctoral period had a transformative effect on her academic career trajectory, enabling her to overcome the challenge of insufficient publications and gain the symbolic capital of international academic mobility.

The cases of Min and Wen further emphasize the advantages of conducting postdoctoral research within the doctoral mentor's network. Both were male with similar predoctoral performance and spent comparable time in the USA for their postdoctoral research. However, Min conducted his postdoctoral research in the same institute as his doctoral mentor did, allowing him to engage in 4 years of continuous research closely aligned with his previous research background. As a result, he published two highly cited papers during his postdoctoral stage. In contrast, Wen secured his postdoctoral position through global solicitation, experiencing two common dangers for postdocs (Musselin 2004). He had to deal with different risky research topics in changing locations within a 5-year period. Although he remained productive throughout, he struggled to obtain satisfactory offers for independent research positions in China due to the lack of continuous accumulation in a focused field. Ultimately, he returned as a co-principal investigator (PI) without autonomy in research direction and student recruitment.

Comparing the cases of Jin, Han, Min, and Wen, it demonstrates that conducting postdoctoral research within the doctoral mentor's network ensures a relatively stable postdoctoral experience and allows for continuous research on suitable topics, playing a critical role in their eventual career success. While existing literature has shown the influence of inheritable international networks in shaping mobility patterns (Melin 2004), our study also reveals its impact on shaping mobility experiences in overcoming the two dangers of risky research topics and contract renewability.

Their continuous interactions with doctoral mentors provide solutions to the third danger, decreased domestic social capital. By taking postdoctoral positions within their doctoral mentors' networks, Jin, Han, and Min were able to maintain better contact and collaboration with their domestic mentors, thereby retaining their strong connections. Meanwhile, their doctoral mentors acted as teachers, providing them career advice and increasing communication. For example, Han stated that she stayed in close contact with her domestic

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mentor throughout her entire postdoctoral period, particularly 'at the critical stage of job hunting' (Hanah, second interview).

At the end of the postdoctoral stage, the advantage of domestic job security due to mentor eminence was evident in the cases of Jin, Han, and Wen. Despite facing unstable postdoctoral appointments and changing research fields, Wen did not express much concern about academic employment, ultimately obtaining a co-PI position arranged by his doctoral mentor. Han also acknowledged her mentor's role during her job-hunting process: 'Domestic *Guanxi* has some influences during the job-hunting process, in other words, the doctoral (ascriptive) background. Publications are the most important, though'. (Han, second interview)

The benefits of mentor eminence in career opportunities align with previous research, but our study reveals that it also leads to further expectations of privileged status upon returning. Jin secured his tenure-track position in an organization closely connected to his doctoral mentor: 'Many things in my career are under the doctoral mentors' arrangement. If I don't have these domestic connections, I may encounter a lot of difficulties in a strange research institute upon returning, such as personnel relation and research resource issues.' (Jin, second interview)

Furthermore, our findings indicate that job security advantages significantly influence decisions regarding postdoctoral international mobility. For instance, both Lang and Rong decided to quit academia due to a lack of academic interest and the highly competitive nature of academic life. However, their perceptions of postdoctoral international mobility differed. Rong chose to conduct postdoctoral research in China to avoid the uncertainty of waiting for employment for many years, while Lang never considered academic job placement as problematic due to his mentor's strong *Guanxi* for academic placement.

Surprisingly, Jin and Han, who emerged as the most successful postdocs in this study, did not express strong aspirations for an academic career during their doctoral education. Instead, they both highlighted the notable influence of their mentors on their decisions to pursue an academic career and conduct postdoctoral research abroad. Initially, Jin's career plan involved becoming a government administrator, but his mentor's intense criticism of this plan led him to reconsider his path. Similarly, Han's original plan was rather vague, described only as 'a good job' (Han, first interview). However, her eventual decision to pursue an academic career was influenced by the lack of satisfactory opportunities in the industrial sector, further reinforced by her mentor's strong expectations for his students to become scholars (Han, first interview). Notably, the mentors of Jin, Han, and Lang exhibited a stronger tendency to guide their mentees towards academic career paths and exerted significant influences on their career decisions.

Previous literature has explained this tendency, suggesting that mentees' academic career decisions are often influenced by their mentors' self-justification of their own career choices and a desire to expand their academic influence (Blackburn, Chapman, and Cameron 1981). The Chinese cultural context of Confucianism and *Shimen* can further amplify this influence, leading PhD students to closely follow their mentors' advice. In this regard, PhD students are often regarded as 'the mentor's child' (Han, second interview), and their career trajectories can largely be 'under the doctoral mentors' arrangement' (Jin, second interview). However, this situation may result in misallocation of resources. For instance, Lang, who lacked academic motivation, pursued overseas postdoctoral positions due to his mentor's tendency to 'send his mentees abroad' and place them in academia through 'strong *Guanxi*' (Lang, second interview). This may lead to passionate postdocs, like Yan, facing limited opportunities.

5.2 Domestic postdocs struggling for academic careers: Yan and Ben

Despite having an impressive predoctoral publication record, Yan chose to pursue postdoctoral research in China due to unfinished research commitments and the lack of attractive overseas postdoctoral opportunities (Yan, first interview). Surprisingly, despite his outstanding publication profile, he did not receive satisfying offers abroad, potentially due to the limited international academic network of his doctoral mentor, who lacked long-term overseas experiences (as shown in Table 3).

Yan was selected into the prestigious Postdoctoral Innovative Talents Support Plan, which supports exceptional postdocs in China. Despite actively participating in ten international conferences to build his academic network and maintaining a high level of productivity, he faced challenges in securing independent research positions at top elite universities due to the lack of long-term international academic mobility. In contrast to the successes of Jin, Han, and Min, Yan's experience highlights the critical role of international academic mobility in gaining cumulative advantages within the current Chinese academic labour market context. Consequently, many outstanding domestic postdocs selected for the same programme eventually left academia: 'As it was claimed at the beginning of the programme, we were all supposed to become elite scholars in different research fields in the future, or at least remain in academia. However, so far, many of us have taken industrial jobs or other positions outside of academia'. (Yan, first interview)

According to Yan, even the most exceptional domestic postdocs face considerable challenges in securing promising academic career prospects, a situation that becomes even more daunting for those without extraordinary publications. Ben's research output during his doctoral education was comparable to Han's, but he did not experience the same 'turn-over' opportunity as a domestic postdoc. As revealed in follow-up interviews, Ben had a strong passion for scientific research, describing it as 'fighting to the end' (Ben, second interview). He decided to continue his postdoctoral research in his doctoral mentor's group with plans to seek overseas opportunities later. However, unexpected circumstances arose when his mentor assumed an administrative position, shifting focus from academic research to administrative affairs. Consequently, Ben was burdened with various management tasks, including lab safety, public service, and administrative work, leading to a lack of academic publications during the initial 2 years of his postdoctoral research (Ben, second interview).

It can be inferred that Ben played the role of an associate supervisor within the research group, a common practice among natural science postdocs in China (Song and Yang 2023). Guided by Chinese sociocultural norms, particularly

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the Confucian value of collective harmony and the *Shimen* culture, Chinese postdocs seldom express negative attitudes toward their supervisors' demands and often take on additional responsibilities related to student supervision and office affairs compared to their international counterparts. As a result, Ben found himself trapped in a dilemma:

> 'Considering the current policy, I need to go abroad and get published. However, I have several projects in progress, half done, or nearly done. It seems the best choice is remaining in this group to get them published. Without these publications, I don't have the choice to apply for an overseas position.' (Ben, second interview)

Struggling with this dilemma for nearly 5 years as a domestic postdoc, Ben's aspirations for an independent academic career gradually waned. He eventually convinced himself that working as a research associate was a pragmatic choice to continue his commitment to 'fighting to the end' in scientific research.

5.3 Combining dual advantages of domestic and overseas postdocs: the story of zheng

Zheng's case provides a more comprehensive perspective on both domestic and overseas postdoctoral experiences, as he had three postdoctoral fellowships in both China and the USA. During his doctoral education, Zheng's publication record is outstanding, and before embarking on his 3-year postdoctoral research in the USA, he initially spent 2 years as a domestic postdoc in his doctoral mentor's group. This decision was driven by considerations of the risks associated with postdoctoral mobility:

> 'From a utility perspective, I planned to take advantage of the 2-year domestic postdoctoral research so that I could continue my research with continuity for better and faster publications. I knew it was not easy to be productive as an overseas postdoc, as it would require adapting to new research topics and environments… In China, it is better to stay in the same research group for continuous research, and it is possible to get published soon'. (Zheng, first interview)

Zheng successfully maintained productivity during his domestic postdoc stage as intended. However, he encountered a typical issue for overseas postdocs during his first postdoctoral research in the US: his contract could not be renewed after one year due to his postdoctoral mentor's retirement. Consequently, he had to move to another research group, resulting in a shift in research topics and environment.

> 'I didn't know he was going to retire when I accepted this offer. If I knew, I wouldn't have gone there… There's no extra internal information, unless your doctoral mentor knows someone there… The research was interrupted after one year, and it negatively impacted research continuity and output… It may take at least 2 or 3 months, and sometimes half a year, to get used to new research directions and projects.' (Zheng, second interview)

As Zheng spent 3 years in the USA, participating in two postdoctoral programmes in different research teams and locations, his publication speed decreased significantly compared to his domestic research period. This aligns with the discussions on Jin, Han, Min, and Wen's cases, highlighting the significant impact of research continuity on productivity. Learning from this experience, Zheng carefully selected his second overseas postdoctoral programme, choosing a research group with connections to his doctoral mentor, which allowed him to spend two consecutive years there. This choice afforded him the opportunity to establish a settled life and maintain a relatively stable research environment (Zheng, second interview). Such stability was particularly important when the Covid-19 pandemic heightened the sense of uncertainty and anxiety among other mobile students and academics at that time (Spronken-Smith et al. 2023).

Additionally, Zheng's unique career trajectory was influenced by his doctoral mentor's role model effect and career advice. Zheng's doctoral mentor only had two years of international research experience as a postdoc in the UK, but he had been productive in China for several years. During the second interview, Zheng mentioned his mentor's guidance regarding his career plans: 'If you have certain academic career aspirations, he will help you with a detailed plan to achieve it… He helped me by discussing future career plans and career development, analysing various factors, and recommending suitable positions'. (Zheng, first and second interview)
Despite encountering challenges during his overseas postdoctoral research, Zheng secured an advanced academic position at a prestigious Chinese university within his doctoral mentor's network after 3 years in the USA. Zheng's career strategy focused on risk management proved to be effective, allowing him to accumulate impressive publications as a domestic postdoc and gain the symbolic capital of international mobility as an overseas postdoc. Additionally, he maintained close contact with his doctoral mentor, resulting in five co-authored papers during Zheng's 3-year postdoctoral research in the USA.

6. Discussions and conclusions

Based on a 4-year longitudinal research on ten postdocs, in this paper, we analysed the interactions between postdoctoral international mobility and the doctoral mentor's role. Beyond the exposure to opportunities, we found that doctoral mentors with long-term overseas research experiences have comprehensive influences on their mentees' access, decisions, experiences, and outcomes of postdoctoral mobility through multiple roles. As teachers, they act as role models and provide career advice for international mobility. As sponsors, they influence not only the opportunities but also the experiences of postdoctoral academic mobility, leading to privileged treatment and research continuity for some overseas postdocs. Moreover, postdocs can collaborate with their domestic mentors, particularly when their research topics align or when their postdoctoral networks are closely connected to their doctoral mentors. These enable postdocs to maintain domestic social capital by staying in contact and collaboration with their doctoral mentors and the associated domestic networks.

Consequently, for PhD graduates supervised by eminent doctoral mentors with extensive overseas experience, managing national social capital and precarity during international mobility becomes easier. This puts them in an advantageous position compared to their peers, especially in the context of the imperative for mobility. On one hand, the ascriptive factor of doctoral mentor is magnified via the professional support

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and arrangement in postdoctoral international mobility, thus increasing the inequality among PhD graduates (Courtois and O'Keefe 2024). On the other hand, some young scholars' early careers are closely bound to their doctoral mentors, which may have negative influences on future scholars' inner passion and autonomy in developing academic careers (Lai, Shi, and Lam 2023). Therefore, the winners of the game may not so match the ideal scholars' characteristics of independence, passion, and excellence.

Firstly, these findings can enrich the research on international mobility as a new factor within the mechanism of cumulative advantages, and extend our understanding on the implicit and long-term consequences of the norm of international mobility. Previous literature has illuminated how international mobility can exacerbate disparities rooted in gender, PhD origins, and social status (Bilecen and Van Mol 2017; Schaer, Dahinden, and Toader 2017; Gerhards, Hans, and Drewski 2018; Ivancheva, Lynch, and Keating 2019); however, this paper sheds light on its influence on the doctoral mentor factor, serving as a crucial addition to the existing dialogue. Combining these findings together, it resonates with the concern that the benefits of mobility are mediated by multiple forms of structural inequality and may be no more than a multiplier of existing resources (Courtois and O'Keefe 2024). Therefore, the legitimacy of mobility imperative is further questioned.

Secondly, our study also provides insights into how doctoral mentors exert continuous influence on their mentees' academic careers. Prior research has indicated that mentors' influence on mentees' career advancement can manifest through diverse avenues, including scientific output and social capital (Cable and Murray 1999; Heinisch and Buenstorf 2018; Rose and Shekhar 2023). Our investigation reveals that, within an academic landscape where international mobility is a normative prerequisite, mentors play a pivotal role in paving the way for their mentees to secure prestigious academic positions through their professional guidance and arrangement of postdoctoral international mobility.

Thirdly, compared to the traditional methods that rely on cross-sectional data, we also highlight the method of longitudinal research in this field. More attention should be paid to this methodological consideration, as the disparities in both opportunities and outcomes are inherently intertwined, rather than occurring in isolation (Bilecen and Van Mol 2017; Courtois and O'Keefe 2024).

Based on these findings, we call for more research on the generalization of our conclusions, considering the unique cultural context of *Guanxi* and *Shimen* in this study. Different countries may adopt diverse local solutions, principles, and practices to address the global norm of international academic mobility (Musselin 2004). Therefore, it is essential to explore whether similar patterns exist in the national academic labour markets of other countries and to what extent they are similar to or different from the Chinese pattern.

Meanwhile, there are some other important external factors that can significantly impact postdoctoral international mobility. For example, the Covid-19 pandemic mentioned in this paper greatly changed the choices and experiences of international academic mobility (Xu, Shen, and Chen 2022; Spronken-Smith et al. 2023), though in this paper most postdocs' overseas research life had ended or was nearing its end in 2020 (only Zheng was an exception) and thus were largely exempt from this influence. In addition, geopolitical changes and the rise of populism have produced profound changes in international academic mobility during the past few years. For instance, in the past few years the escalating US–China tech war may be prompting the USA to adopt a more cautious stance towards accepting international postdoctoral fellows in science and engineering fields (Kennedy 2023; Mok, Shen, and Gu 2024). In contrast to 2016 (when the ten postdoctoral researchers in this study were just starting their postdoctoral stage), the opportunities for international academic mobility are becoming increasingly limited today. It requires timely observations and analysis on whether the ascriptive factors have been further amplified upon the decreased mobility opportunities, and whether there are emerging changes of attitude towards international mobility within Chinese academia.

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Data availability

The data underlying this article will be shared on reasonable request to the corresponding author.

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